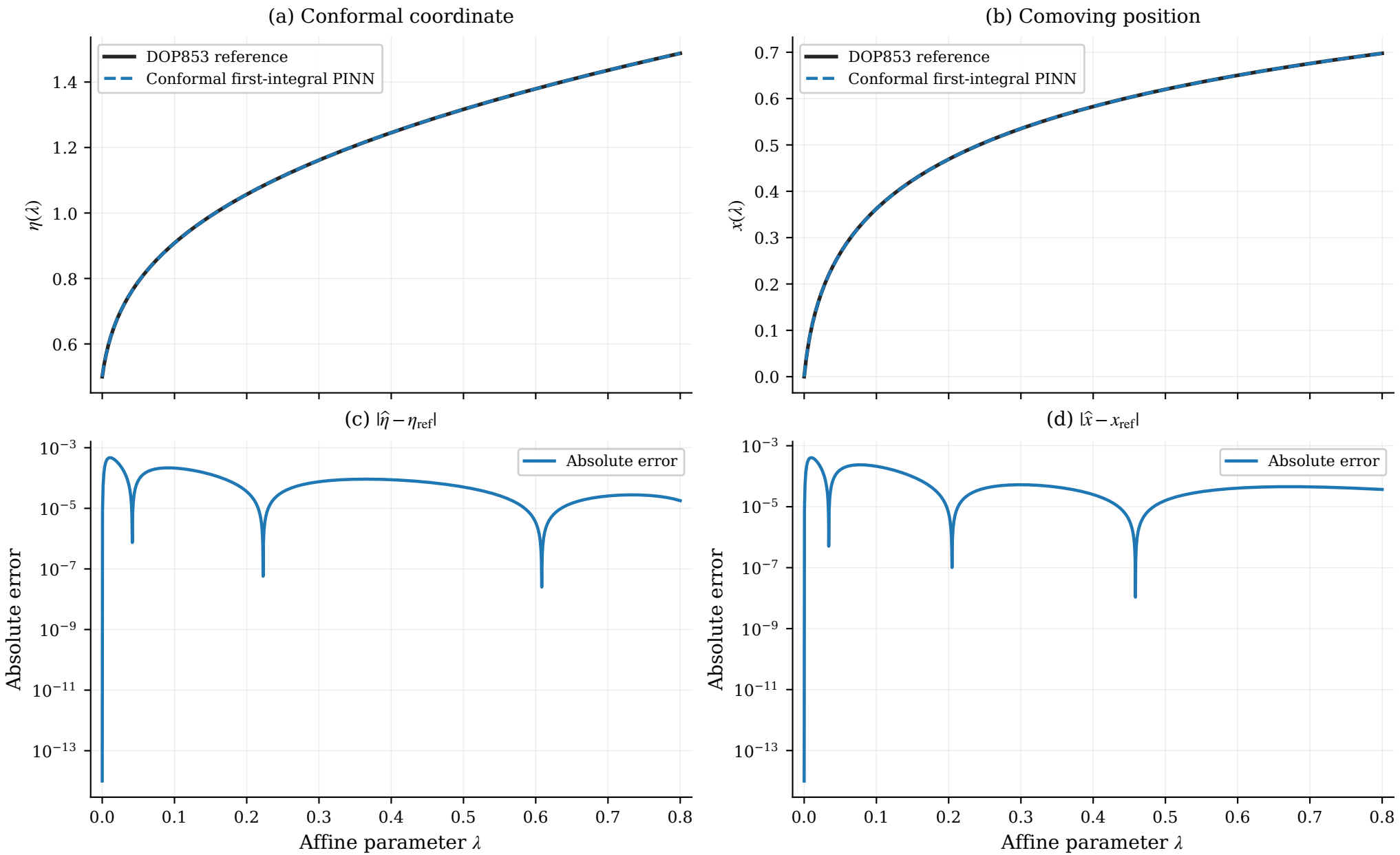
(a) Conformal coordinate
DOP853 reference
Conformal first-integral PINN
$\eta(\lambda)$
(b) Comoving position
$x(\lambda)$
(c) $|\hat{\eta} - \eta_{\mathrm{ref}}|$
Absolute error
(d) $|\hat{x} - x_{\mathrm{ref}}|$
Absolute error
Affine parameter λ
Affine parameter λ